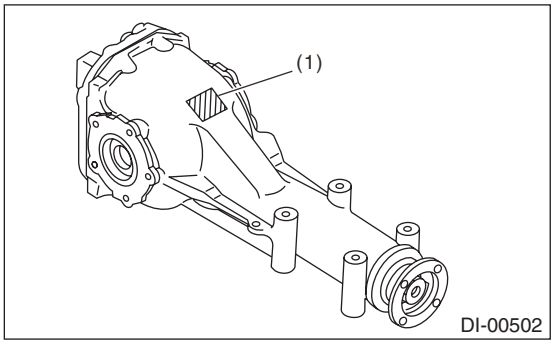
- **Identification**

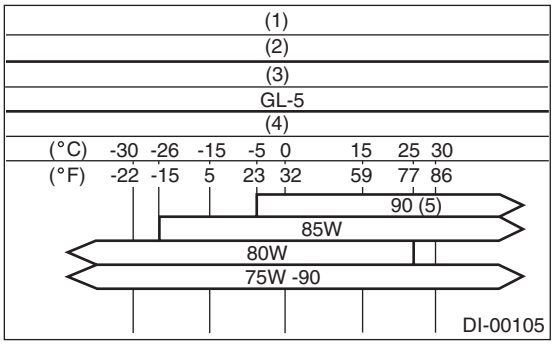


(1) Type (paint)

- **Rear differential gear oil**

Recommended oil

CAUTION:
Each oil manufacturer uses different basic oil and additives. Thus, do not mix two or more brands.



(1) Item
(2) Rear differential gear oil
(3) API standard
(4) SAE viscosity No. and applicable temperature
(5) STI model

1. SERVICE DATA

Drive pinion bearing preload (For new bearing)	Measure with spring mesurement. (Measured from the companion flange bolt) N (kgf, lbf)	T-type (Except for STI model)	17.7 — 38.8 (1.8 — 4.0, 4.0 — 8.7)
		T-type (STI model)	24.1 — 38.6 (2.5 — 3.9, 5.4 — 8.7)
	Measure with torque wrench N·m (kgf-m, ft-lb)	T-type (Except for STI model)	0.67 — 1.47 (0.07 — 0.15, 0.49 — 1.08)
		T-type (STI model)	0.98 — 1.57 (0.10 — 0.16, 0.7 — 1.16)
Side gear backlash		mm (in)	0.10 — 0.20 (0.004 — 0.008)
Side bearing standard width		mm (in)	20.00 (0.7874)
Hypoid driven gear to drive pinion backlash		mm (in)	0.10 — 0.20 (0.004 — 0.008)
Driven gear rear face runout limit		mm (in)	0.05 (0.0020) or less

2. ADJUSTING PARTS

Except for STI model

Drive pinion bearing preload (For new bearing)	Measure with spring measurement. (Measured from the companion flange bolt) N (kgf, lbf)	17.7 — 38.8 (1.8 — 4.0, 4.0 — 8.7)	
	Measure with torque wrench N·m (kgf-m, ft-lb)	0.67 — 1.47 (0.07 — 0.15, 0.49 — 1.08)	
Preload adjusting spacer		Part No.	Length mm (in)
		383695201	56.2 (2.213)
		383695202	56.4 (2.220)
		383695203	56.6 (2.228)
		383695204	56.8 (2.236)
		383695205	57.0 (2.244)
		383695206	57.2 (2.252)
Preload adjusting washer		Part No.	Length mm (in)
		383705200	2.59 (0.1020)
		383715200	2.57 (0.1012)
		383725200	2.55 (0.1004)
		383735200	2.53 (0.0996)
		383745200	2.51 (0.0988)
		383755200	2.49 (0.0980)
		383765200	2.47 (0.0972)
		383775200	2.45 (0.0965)
		383785200	2.43 (0.0957)
		383795200	2.41 (0.0949)
		383805200	2.39 (0.0941)
		383815200	2.37 (0.0933)
		383825200	2.35 (0.0925)
		383835200	2.33 (0.0917)
		383845200	2.31 (0.0909)
Pinion height adjusting washer		Part No.	Thickness mm (in)
		383495200	3.09 (0.1217)
		383505200	3.12 (0.1228)
		383515200	3.15 (0.1240)
		383525200	3.18 (0.1252)
		383535200	3.21 (0.1264)
		383545200	3.24 (0.1276)
		383555200	3.27 (0.1287)
		383565200	3.30 (0.1299)
		383575200	3.33 (0.1311)
		383585200	3.36 (0.1323)
		383595200	3.39 (0.1335)
		383605200	3.42 (0.1346)
		383615200	3.45 (0.1358)
		383625200	3.48 (0.1370)
		383635200	3.51 (0.1382)
		383645200	3.54 (0.1394)
		383655200	3.57 (0.1406)
		383665200	3.60 (0.1417)
		383675200	3.63 (0.1429)
		383685200	3.66 (0.1441)

Side gear backlash mm (in)		0.1 — 0.2 (0.004 — 0.008)	
Side gear thrust washer (Model without LSD)		Part No.	Thickness mm (in)
		383445201	0.75 — 0.80 (0.0295 — 0.0315)
		383445202	0.80 — 0.85 (0.0315 — 0.0335)
		383445203	0.85 — 0.90 (0.0335 — 0.0354)
Side bearing standard width mm (in)		—	20.00 (0.7874)
Side bearing retainer shim		Part No.	Thickness mm (in)
		383475201	0.20 (0.0079)
		383475202	0.25 (0.0098)
		383475203	0.30 (0.0118)
		383475204	0.40 (0.0157)
		383475205	0.50 (0.0197)
Limit mm (in)	Hypoid driven gear to drive pinion backlash	—	0.10 — 0.20 (0.004 — 0.008)
	Driven gear rear face runout limit		0.05 (0.0020)

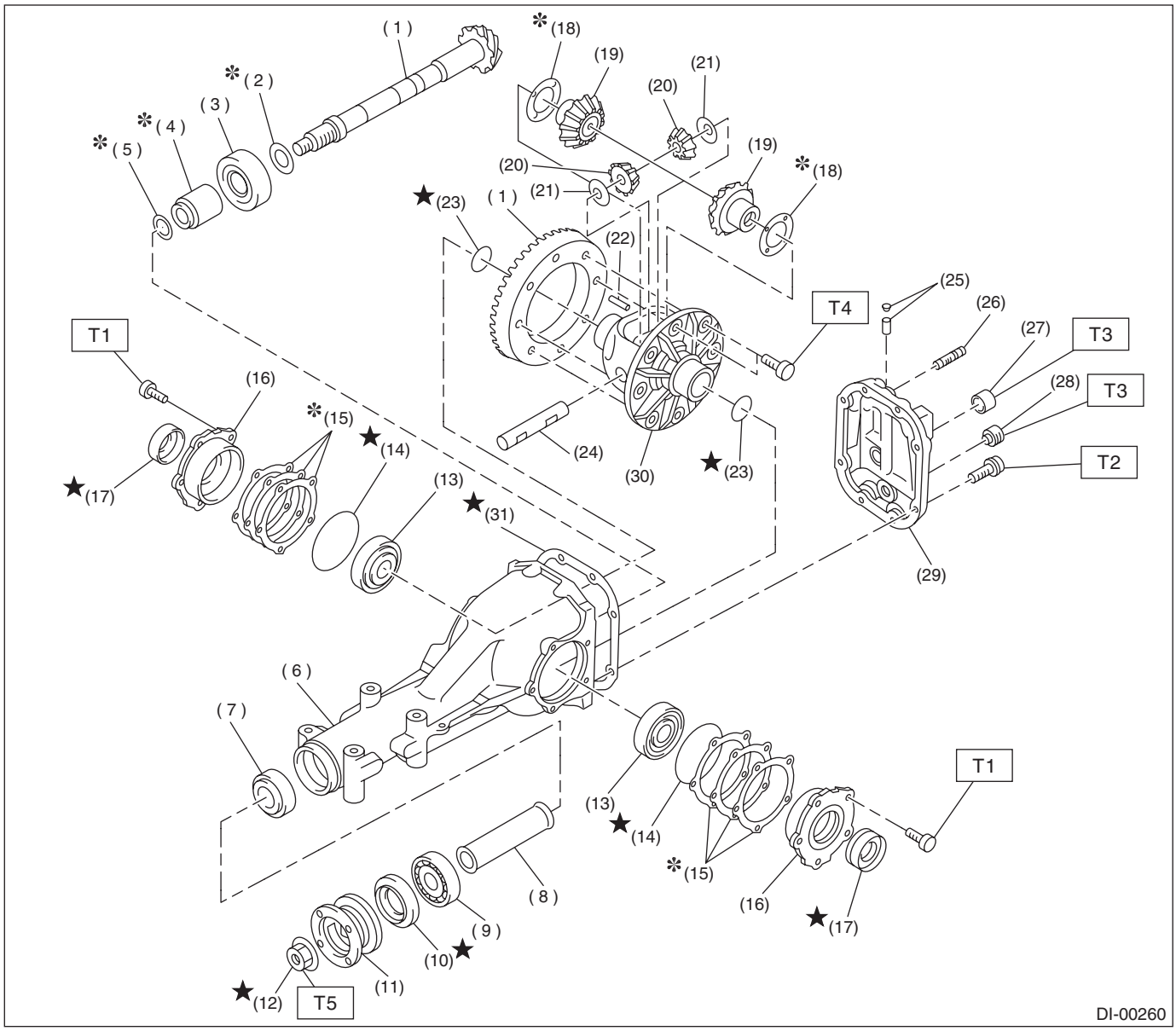
STI model

Drive pinion bearing preload (For new bearing)	Measure with spring measurement. (Measured from the companion flange bolt) N (kgf, lbf)	24.1 — 38.6 (2.5 — 3.9, 5.4 — 8.7)	
	Measure with torque wrench N·m (kgf-m, ft-lb)	0.98 — 1.57 (0.10 — 0.16, 0.72 — 1.16)	
Preload adjusting spacer		Part No.	Length mm (in)
		31454AA130	52.2 (2.055)
		31454AA140	52.4 (2.063)
		31454AA150	52.6 (2.071)
		31454AA160	52.8 (2.079)
		31454AA170	53.0 (2.087)
		31454AA180	53.2 (2.094)
Preload adjusting washer		Part No.	Length mm (in)
		383705200	2.59 (0.1020)
		383715200	2.57 (0.1012)
		383725200	2.55 (0.1004)
		383735200	2.53 (0.0996)
		383745200	2.51 (0.0988)
		383755200	2.49 (0.0980)
		383765200	2.47 (0.0972)
		383775200	2.45 (0.0965)
		383785200	2.43 (0.0957)
		383795200	2.41 (0.0949)
		383805200	2.39 (0.0941)
		383815200	2.37 (0.0933)
		383825200	2.35 (0.0925)
		383835200	2.33 (0.0917)
		383845200	2.31 (0.0909)
Pinion height adjusting washer		Part No.	Length mm (in)
		38336AA230	3.09 (0.1217)
		38336AA240	3.12 (0.1228)
		38336AA250	3.15 (0.1240)
		38336AA260	3.18 (0.1252)
		38336AA270	3.21 (0.1264)
		38336AA280	3.24 (0.1276)
		38336AA290	3.27 (0.1287)
		38336AA300	3.30 (0.1299)
		38336AA310	3.33 (0.1311)
		38336AA320	3.36 (0.1323)
		38336AA330	3.39 (0.1335)
		38336AA340	3.42 (0.1346)
		38336AA350	3.45 (0.1358)
		38336AA360	3.48 (0.1370)
		38336AA370	3.51 (0.1382)
		38336AA380	3.54 (0.1394)
		38336AA390	3.57 (0.1406)
		38336AA400	3.60 (0.1417)
		38336AA410	3.63 (0.1429)
		38336AA420	3.66 (0.1441)
Side bearing standard width	mm (in)	20.00 (0.7874)	

		Part No.	Thickness mm (in)
Side bearing retainer shim		383475201	0.20 (0.0079)
		383475202	0.25 (0.0098)
		383475203	0.30 (0.0118)
		383475204	0.40 (0.0157)
		383475205	0.50 (0.0197)
Limit mm (in)	Hypoid driven gear to drive pinion backlash	—	0.10 — 0.20 (0.004 — 0.008)
	Driven gear rear face runout limit		0.05 (0.0020)

B: COMPONENT

1. REAR DIFFERENTIAL WITHOUT LSD

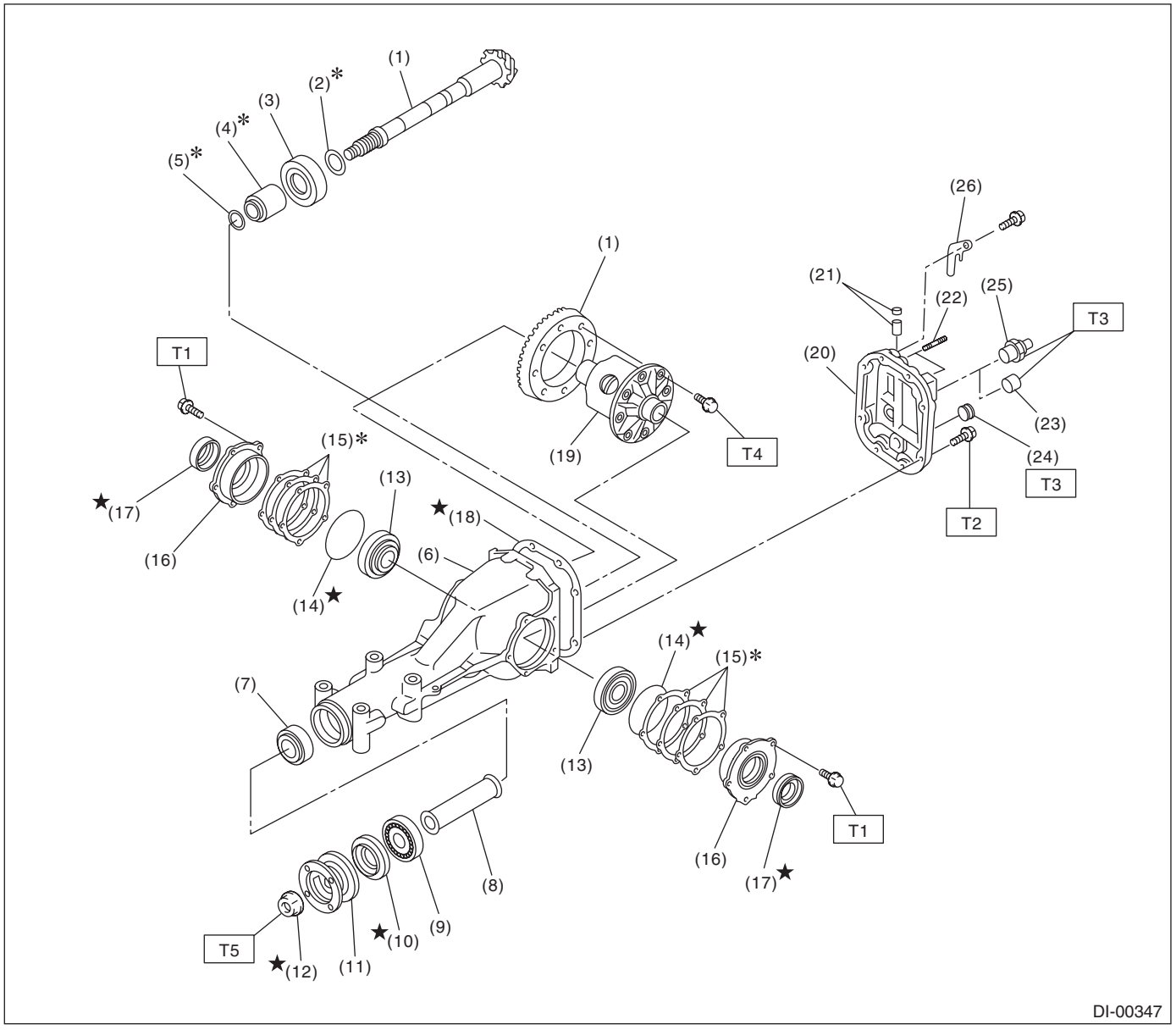


(1)	Hypoid driven gear and drive pinion set	(13)	Side bearing	(26)	Stud bolt
(2)	Pinion height adjusting washer	(14)	O-ring	(27)	Oil filler plug
(3)	Rear bearing	(15)	Side bearing retainer shim	(28)	Oil drain plug
(4)	Bearing preload adjusting spacer	(16)	Side bearing retainer	(29)	Rear cover
(5)	Bearing preload adjusting washer	(17)	Side oil seal	(30)	Differential case
(6)	Differential carrier	(18)	Side gear thrust washer	(31)	Gasket
(7)	Front bearing	(19)	Side gear		
(8)	Spacer	(20)	Pinion mate gear		
(9)	Pilot bearing	(21)	Pinion mate gear washer		
(10)	Front oil seal	(22)	Pinion shaft lock pin		
(11)	Companion flange	(23)	Snap ring		
(12)	Self-locking nut	(24)	Pinion mate shaft		
		(25)	Air breather cap		

Tightening torque:N·m (kgf-m, ft-lb)

T1: 10.5 (1.1, 7.7)
T2: 29.5 (3.0, 21.8)
T3: 49 (5.0, 36.2)
T4: 103 (10.5, 76)
T5: 181.5 (18.5, 134)

2. REAR DIFFERENTIAL WITH LSD

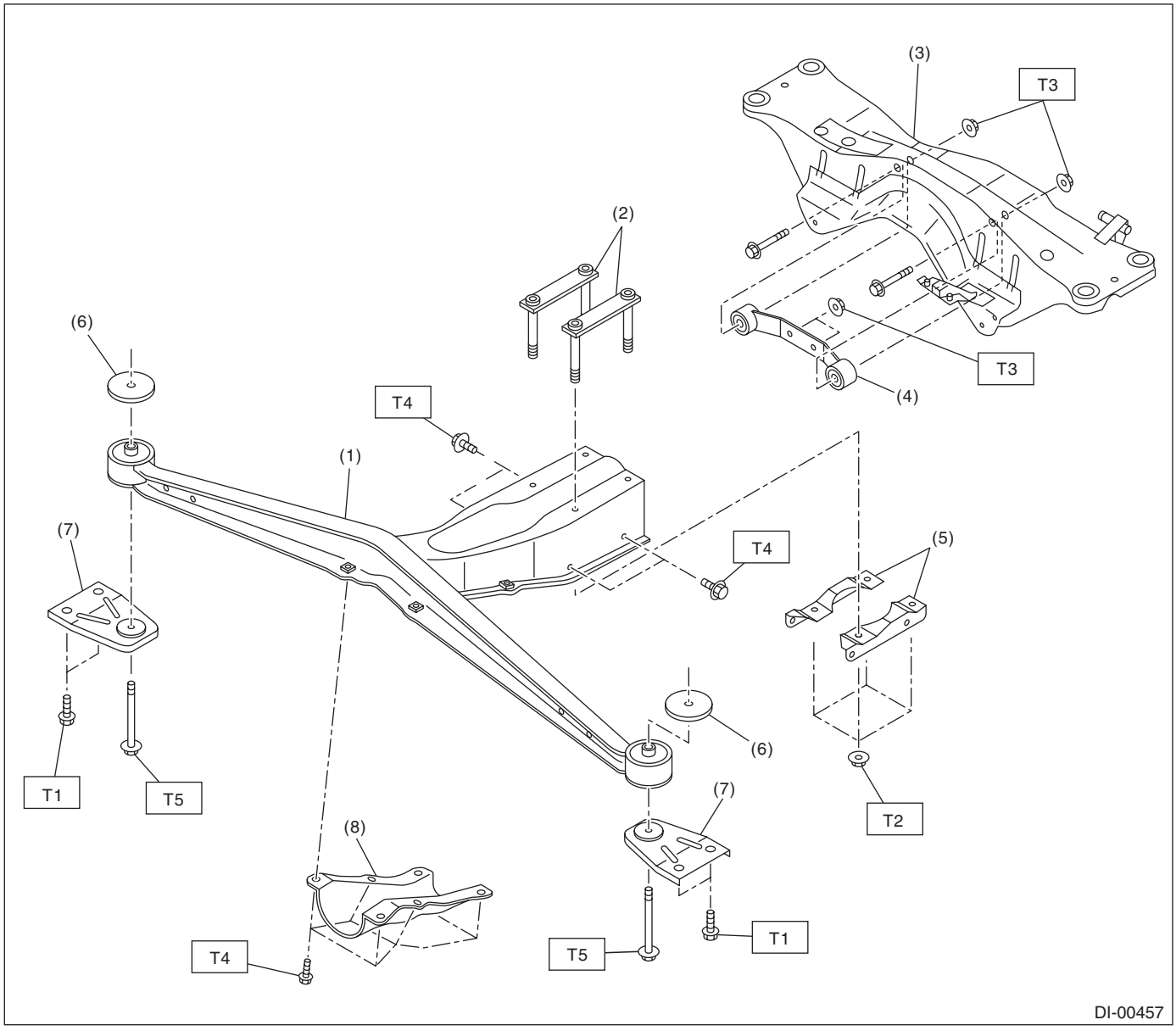


(1) Hypoid driven gear and drive pinion set
(2) Pinion height adjusting washer
(3) Rear bearing
(4) Bearing preload adjusting spacer
(5) Bearing preload adjusting washer
(6) Differential carrier
(7) Front bearing
(8) Collar
(9) Pilot bearing
(10) Front oil seal
(11) Companion flange
(12) Self-locking nut
(13) Side bearing
(14) O-ring
(15) Side bearing retainer shim
(16) Side bearing retainer
(17) Side oil seal
(18) Gasket
(19) Differential case (Viscous coupling type) (Except for STI model)
Differential case (Mechanical type) (STI model)
(20) Rear cover
(21) Air breather cap
(22) Stud bolt
(23) Oil filler plug
(24) Oil drain plug
(25) Oil filler plug (Rear differential oil temperature switch) (STI model)
(26) Stay ground (STI model)

Tightening torque:N·m (kgf-m, ft-lb)

T1:	***10.5 (1.1, 7.7)***
T2:	***29.5 (3.0, 21.8) (except for STI model)***
	44 (4.5, 32.5) (STI model)
T3:	***49 (5.0, 36.2)***
T4:	***103 (10.5, 76)***
T5:	***181.5 (18.5, 134)***

3. REAR DIFFERENTIAL MOUNTING SYSTEM



(1)	Differential front member	(5)	Differential mount lower bracket	***Tightening torque:N·m (kgf-m, ft-lb)***
(2)	Plate	(6)	Stopper	***T1: 33 (3.4, 24.3)***
(3)	Crossmember	(7)	Differential mount bracket	***T2: 50 (5.1, 36.9)***
(4)	Differential rear member	(8)	Differential mount front cover	***T3: 70 (7.1, 51.6)***
				T4: 90 (9.2, 66.4)
				T5: 100 (10.2, 73.8)

C: CAUTION

- Wear appropriate work clothing, including a cap, protective goggles and protective shoes when performing any work.
- Remove contamination including dirt and corrosion before removal, installation or disassembly.
- Keep the disassembled parts in order and protect them from dust and dirt.
- Before removal, installation or disassembly, be sure to clarify the failure. Avoid unnecessary removal, installation, disassembly and replacement.
- Vehicle components are extremely hot after driving. Be wary of receiving burns from heated parts.
- Use SUBARU genuine gear oil, grease or the equivalent. Do not mix gear oil, grease, etc. of different grades or manufacturers.
- Be sure to tighten fasteners including bolts and nuts to the specified torque.
- Place shop jacks or rigid racks at the specified points.
- Apply gear oil onto sliding or revolving surfaces before installation.
- Before installing the O-ring or snap ring, apply a sufficient amount of gear oil to avoid damage and deformation.
- Before securing a part on a vise, place cushioning material such as wood blocks, aluminum plate, or cloth between the part and the vise.
- Avoid damaging the mating surface of the case.

D: PREPARATION TOOL

1. SPECIAL TOOL

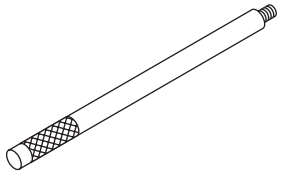
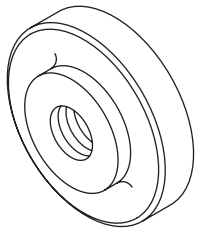
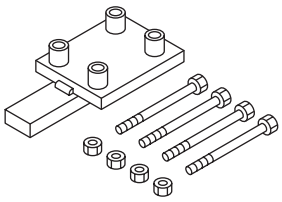
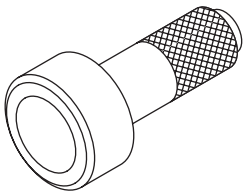
ILLUSTRATION	TOOL NUMBER	DESCRIPTION	REMARKS
 ST-398477701	398477701	HANDLE	Used for installing the front and rear bearing cones.
 ST-398477702	398477702	DRIFT	Used for press-fitting the bearing race (front) of the differential carrier.
 ST-398217700	398217700	ATTACHMENT SET	Stand for rear differential carrier disassembly and assembly.
 ST-498447120	498447120	INSTALLER	Used for installing the front oil seal.

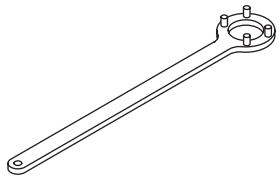
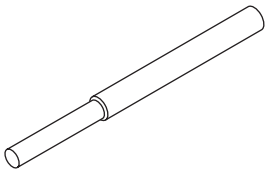
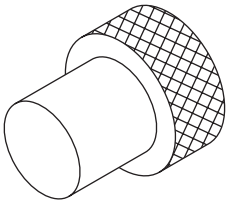
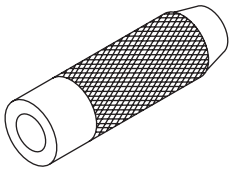
ILLUSTRATION	TOOL NUMBER	DESCRIPTION	REMARKS
 ST-498427200	498427200	FLANGE WRENCH	• Used for preventing rotation of companion flange when loosening and tightening self-lock nut. • For models except for STI model.
 ST-398467700	398467700	DRIFT	Used for removing pinion, pilot bearing and front bearing cone.
 ST-399780104	399780104	WEIGHT	Used for installing the front bearing cone and the pilot bearing companion flange.
 ST-899580100	899580100	INSTALLER	Used for press-fitting the front bearing cone and pilot bearing.

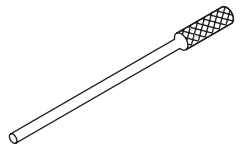
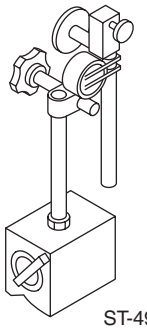
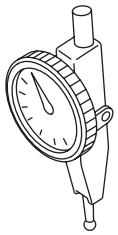
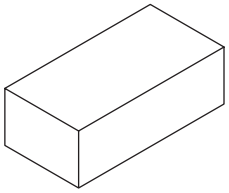
ILLUSTRATION	TOOL NUMBER	DESCRIPTION	REMARKS
 ST-899904100	899904100	STRAIGHT PIN REMOVER	Used for driving out differential pinion shaft lock pin.
 ST-498247001	498247001	MAGNET BASE	• Used for measuring backlash between side gear and pinion, and hypoid gear. • Used together with the DIAL GAUGE (498247100).
 ST-498247100	498247100	DIAL GAUGE	• Used for measuring backlash between side gear and pinion, and hypoid gear. • Used together with the MAGNET BASE (498247001).
 ST-398507704	398507704	BLOCK	Used for adjusting pinion height and preload.

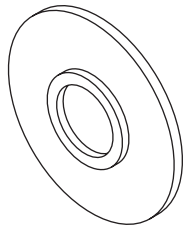
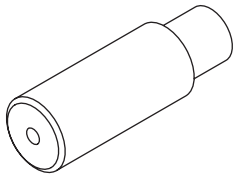
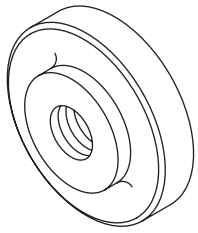
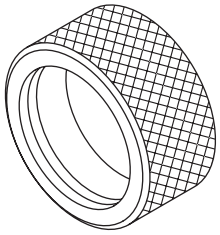
ILLUSTRATION	TOOL NUMBER	DESCRIPTION	REMARKS
 ST-398177700	398177700	INSTALLER	Used for installing the rear bearing cone.
 ST-398457700	398457700	ATTACHMENT	Used for removing the side bearing retainer.
 ST-398477703	398477703	DRIFT 2	Used for press-fitting bearing race (rear) of differential carrier.
 ST-398437700	398437700	DRIFT	Used for installing the side oil seal.

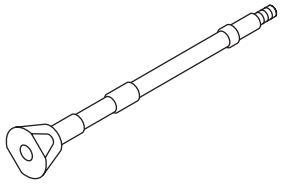
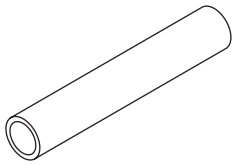
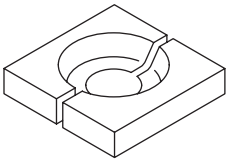
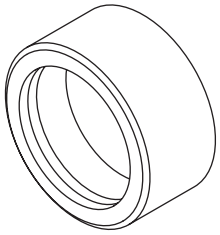
ILLUSTRATION	TOOL NUMBER	DESCRIPTION	REMARKS
 ST-398507702	398507702	DUMMY SHAFT	Used for adjusting pinion height and preload.
 ST-398507703	398507703	DUMMY COLLAR	Used for adjusting pinion height and preload.
 ST-398517700	398517700	REPLACER	Used for removing rear bearing cone.
 ST-398487700	398487700	DRIFT	Used for press-fitting side bearing cone.

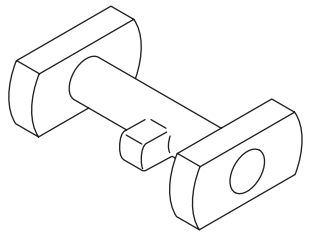
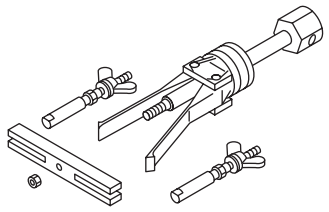
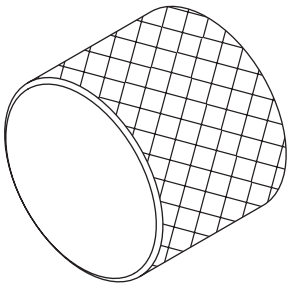
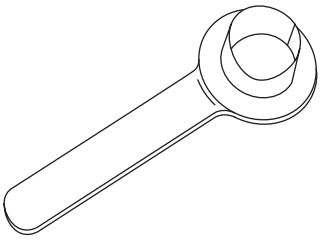
ILLUSTRATION	TOOL NUMBER	DESCRIPTION	REMARKS
 ST-398507701	398507701	DIFFERENTIAL CARRIER GAUGE	Used for adjusting pinion height.
 ST-398527700	398527700	PULLER ASSY	• Used for removing front oil seal. • Used for removing side bearing cup.
 ST-398227700	398227700	WEIGHT	Used for installing side bearing.
 ST28099PA090	28099PA090	OIL SEAL PROTECTOR	• Used for installing the rear drive shaft to the rear differential. • For oil seal protection

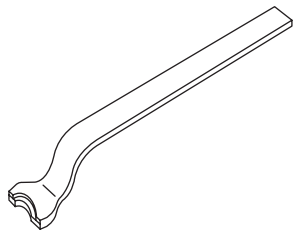
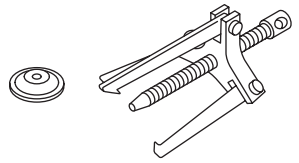
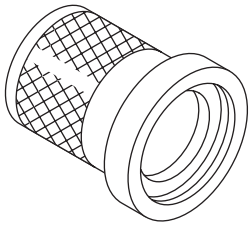
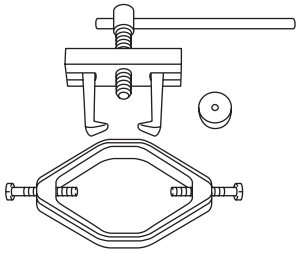
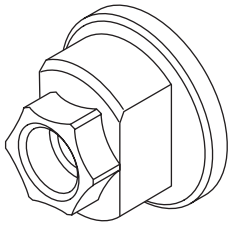
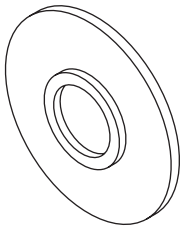
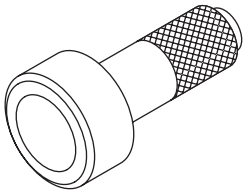
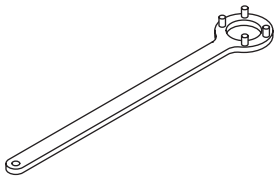
ILLUSTRATION	TOOL NUMBER	DESCRIPTION	REMARKS
 ST28099PA100	28099PA100	DRIVE SHAFT REMOVER	Used for removing the rear drive shaft from rear differential.
 ST-399703600	399703600	PULLER ASSY	Used for removing companion flange.
 ST-899874100	899874100	INSTALLER	Used for installing the companion flange.
 ST18759AA000	18759AA000	PULLER ASSY	Used for removing the differential side bearing cone.

ILLUSTRATION	TOOL NUMBER	DESCRIPTION	REMARKS
 ST-498937110	498937110	HOLDER	• Used for installing the pilot bearing. • For STI model.
 ST18674AA000	18674AA000	INSTALLER	• Used for installing the rear bearing cone. • For STI model.
 ST-398417700	398417700	DRIFT	Used for installing side bearing race.
 ST18633AA000	18633AA000	WRENCH COMPL	• Used for preventing rotation of companion flange when loosening and tightening self-lock nut. • For STI model.

2. GENERAL TOOL

TOOL NAME	REMARKS
Transmission jack	Used for assembly/disassembly of the rear differential.
Puller	Used for removing the side bearing retainer.
Thickness gauge	Used for measuring clearance.